

Nathan R.T. Lesnevich

PERSONAL INFORMATION

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ACADEMIC POSITIONS

2024-CURRENT Postdoctoral Fellow, Oklahoma State University

EDUCATION

MAY 2024	Ph.D in Mathematics, Washington University in St. Louis (WUSTL), St. Louis MO Thesis: Splines on Cayley Graphs of the Symmetric Group. Advisors: Assoc. Prof. Martha Precup Prof. John Shareshian
MAY 2021	A.M. in Mathematics, Washington University in St. Louis (WUSTL), St. Louis MO
MAY 2019	B.S. In Mathematics (<i>magna cum laude</i>), Bucknell University , Lewisburg PA Pure Mathematics concentration, additional fulfillment of B.A. in Computer Science. Thesis: Positivity Among P -Partition Generating Functions Advisor: Prof. Peter McNamara

PUBLICATIONS

7. J. Boretsky, A. Cornejo, R. Hodges, P. Horn, N. Lesnevich, T. McAllister. *Combarability in Bruhat orders*. (2026). [arxiv:2601.21158]
6. N.R.T. Lesnevich. *Signed permutations and degree-one dot action representations for types B and C*. submitted (2025). [arXiv:2511.13913]
5. J. Lee, N. Lesnevich, and M. Precup. *Restricted inversion polynomials*. submitted (2025). [arXiv:2511.05676]
4. B. Adenbaum, E. Barnard, M. Hlavacek, B. Kagy, N. R. T. Lesnevich, G. D. Nasr, and K. Waddle, *The poset of maximal tubings of the cycle graph is a lattice*. submitted (2025). [arXiv:2510.09429]
3. N.R.T Lesnevich, *Splines on Cayley graphs of the symmetric group*. Forum of Mathematics, Sigma 13 (2025): e96. [arxiv:2401.14380]
2. N.R.T. Lesnevich, *Hook-shape immanant characters from Stanley-Stembridge characters*. Algebraic Combinatorics, Volume 7 (2024) no. 1, pp. 137-157. [arXiv:2304.05285].
1. N.R.T. Lesnevich and P.R.W. McNamara, *Positivity among P -partition generating functions*. Annals of Combinatorics. 26, 171-204 (2022). [arxiv:2006.10087]

GRANTS

2022-3	Prepared and received NSF conference grant DMS-2245927 (PI: Martha Precup) <i>Graduate Student Combinatorics Conference 2023</i> Co-written with small group of other graduate students
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TEACHING

FALL 2024-CURRENT	Instructor, Oklahoma State University Dept. of Mathematics <i>Differential Equations, Calculus II, Combinatorics</i>
FALL 2023	Instructor, Graduate Reading Program, WUSTL Dept. of Mathematics <i>Representation Theory</i>
2020-2023	Graduate Teaching Assistant, WUSTL Dept. of Mathematics <i>Discussion Section Leading, Grading</i> Calculus III (FL20,SP23) and II (SP21), and Differential Equations (FL21-FL22).
SUMMER 2022	Finite Math (Math 220), WUSTL Dept. of Mathematics <i>Sole instructor of record.</i>

SPRING 2022	Co-Instructor, Graduate Reading Program, WUSTL Dept. of Mathematics <i>Lie Algebras</i>
FALL 2021	Co-Instructor, Graduate Reading Program, WUSTL Dept. of Mathematics <i>The Art of Counting Lattice Points: Introduction to Ehrhart Theory</i>
SPRING 2021	Co-Instructor, Graduate Reading Program, WUSTL Dept. of Mathematics <i>Young Tableaux</i>
2016-2019	Help Session Leader, Bucknell Dept. of Mathematics <i>Group Tutoring, Calculus 1-3</i>

PROGRAMS & OUTREACH

SUMMER 2024	Research advisor, Freiwald Scholars Summer Research Program, WUSTL Department of Mathematics <i>Co-advised several undergraduate research projects</i>
APR 2024	Graduate Student Assistant, Sonia Kovalevsky Day at Washington University in St. Louis <i>Cryptography Activity Organizer</i>
AUG 2021-MAY 2024	Organizer, Graduate Reading Program, WUSTL Dept. of Mathematics <i>Recruitment, Organizational, and Administrative Duties</i>
SEPT 24 2023	Washington University Math Circle Speaker, "Cryptography: The math of making and breaking codes"

SCHOLARSHIPS AND CERTIFICATES

AC. YEAR 2022-2023	Robert H. McDowell Teaching Award. <i>Department graduate teaching award, 1 annual awardee.</i>
AC. YEAR 2021-2022	Brian Blank Teaching Award for 3rd Year Graduate Student. <i>Department graduate award, 1-2 annual awardees.</i>
FALL-SPRING 2020-21	David B. Emmes Endowed Scholarship.
FALL-SPRING 2015-19	John C. Hoover Scholarship for Students in Mathematics
FALL-SPRING 2015-19	Bucknell Presidential Fellowship <i>4 year half-tuition merit scholarship with research mentorship.</i>
FALL-SPRING 2015-19	Bucknell University's Deans List
SUMMER 2018	John C. Hoover Summer Research Funding Recipient
SPRING 2017	Pi Mu Epsilon Mathematics Honors Society Member
SUMMER 2017	Bucknell Program for Undergraduate Research Award Recipient

CONFERENCES AND SEMINARS

Organization

Spring 2025	Special Session on Combinatorics and Lie Theory AMS Spring Central Sectional Meeting <i>University of Kansas</i>
Spring 2023	Graduate Student Combinatorics Conference 2023 Co-wrote and received NSF conference grant DMS-2245927.
2022-2023	WUSTL Combinatorics Seminar <i>Bi-weekly seminar.</i>

Speaking

Fall 2025	The Schubert Seminar (inter-university) <i>Combinatorics for GKM Theory and the Dot Action, Nov 3</i>
Fall 2025	AMS Fall Central Sectional, Special Session on Interactions between Geometry, Combinatorics, and Flag Varieties <i>The Second Cohomology of $\text{Hess}(X, H)$ and the Dot Action in types B/C, Oct 19</i>

<i>Fall 2025</i>	Oklahoma State Lie Groups Seminar <i>GKM Theory and Flag Varieties</i> , Oct 8
<i>Summer 2025</i>	Graduate Research Workshop in Combinatorics, (at IA State) <i>Dihedral Splines</i> , June 15-27
<i>Spring 2025</i>	WashU Combinatorics Seminar <i>The Maximal Tubings of the Cycle Graph and G-inversions</i> , March 17
<i>Fall 2024</i>	Oklahoma State Combinatorial and Commutative Algebra Seminar <i>Splines on Cayley Graphs of the Symmetric Group</i> , Sept 26
<i>Spring 2024</i>	Poster, Enumerative and Algebraic Combinatorics (at UFL) <i>Splines on Cayley Graphs of the Symmetric Group</i> , Feb 25-7
<i>Fall 2023</i>	KU Combinatorics Seminar <i>Splines on Cayley Graphs of the Symmetric Group</i> , Nov 3
<i>Summer 2023</i>	Short Talk, Schubert Conference (at UIUC) <i>Splines on Cayley Graphs of the Symmetric Group</i> , June 20
<i>Spring 2023</i>	Poster, AlGeCom-XXIII <i>Splines on Cayley Graphs of the Symmetric Group</i> , Apr 15
<i>Spring 2023</i>	Graduate Online Combinatorics Colloquium <i>Splines on Cayley Graphs of the Symmetric Group</i> , Mar 8
<i>Spring 2023</i>	Dartmouth Combinatorics Seminar <i>Splines on Cayley Graphs of the Symmetric Group</i> , Feb 21
<i>Spring 2023</i>	WUSTL Combinatorics Seminar <i>Splines on Cayley Graphs of the Symmetric Group</i> , Feb 20
<i>Fall 2022</i>	AMS Fall Eastern Sectional, Special Session on Geometric Aspects of Algebraic Combinatorics. <i>A Proof of an Immanant Conjecture for Hook Partitions</i>
<i>Spring 2022</i>	Short Talk, Graduate Student Combinatorics Conference 2022 <i>Hook-Partition Immanant Characters from Stanley-Stembridge Characters</i>

Recent Conference Attendance

<i>Summer 2025</i>	Graduate Research Workshop in Combinatorics, at Iowa State University
<i>Summer 2024</i>	AMS Mathematics Research Communities: Algebraic Combinatorics, at Beaver Hollow Convention Center
<i>Spring 2024</i>	BIRS Workshop: Homological Perspectives on Splines, at UBC Kelowna
<i>Summer 2023</i>	FPSAC 2023, at UC Davis
<i>Summer 2023</i>	Schubert Calculus Summer School, at UIUC
<i>Spring 2023</i>	8th Lake Michigan Workshop on Combinatorics and Graph Theory, at the University of Notre Dame
<i>Fall 2022</i>	AlGeCom-XXII, at UIUC.
<i>Fall 2022</i>	BIRS: Interactions between Hessenberg Varieties, Chromatic Functions, and LLT Polynomials (Virtual Participant)
<i>Spring 2022</i>	ISM Discovery School: Geometry and Combinatorics of Hessenberg Varieties, at UQAM.